Department of Electrical and Electrronics Engineering

University of Dhaka

Batch - 103, EEE-58

2^{nd} Year 1^{st} Semester

Course: EEE-2109

Title: Signals and Systems

Class Techer: Md. Shahin Parvej

Note by

Md. Galib Ashraf

Session: 2023-2024

Class Roll: SH-042

SYLLABUS

Signals: Classification of Signals, Basic Operations on Signals, Elementary Signals, The Concept of Frequency in Continuous-Time and Discrete-Time Signals, Analog-to-Digital and Digital-to-Analog Conversion.

Systems: Input-Output Description of Systems, Classification of Systems, Properties of Systems, Interconnection of Systems.

Linear Time-Invariant (LTI) Systems: Response of Continuous-Time LTI System and Convolution Integral, Response of a Discrete-Time LTI System and Convolution Sum, Recursive and Non-Recursive Systems, Differential and Difference Equation Representations of LTI Systems, Solution of Differential and Difference Equations, State Variable Descriptions of LTI Systems, Structures for the Realization of LTI Systems.

Fourier Analysis of Signals and Systems: Fourier Representations of Signals, The Fourier series, The Discrete-Time Fourier Series, The Fourier Transform, The Discrete-Time Fourier Transform, Properties of Fourier Representations, Parseval's Theorem, Power Density Spectrum and Energy Density Spectrum.

Applications of Fourier Analysis: Frequency Response of LTI Systems to Complex Exponential Signals, Filtering and Bandwidth, Frequency AnalysisAnalog Electrical Systems, Sampling and Reconstruction of Signal, Convolution and Multiplication with Mixtures of Periodic and Non-periodic Signals, Modulation and Demodulation, Time-Division and Frequency-Division Multiplexing.

The Laplace Transform: Definition, Properties, Convergence of Laplace Transform, Initial-value and Final-Value Theorem, Inverse Laplace Transform, Partial-fraction Expansions, Heaviside Expansion Formula, Solution of Differential Equation, System Transfer Function, System Stability, Electrical Network Analysis.

REFERENCE BOOKS

1. Signals and Systems, 2^{nd} Edition, Simon Haykin and Barry Van Veen, Wiley Inc.
2. Signals and Systems, 2^{nd} Edition, Alan V. Oppenheim, Alan S. Willsky and S. Hamid, Pearson Education.
3. Linear Systems and Signals, 3^{rd} Edition, B.P. Lathi and Roger Green, Oxford University Press.
4. Signals and Systems, 1^{st} Edition, Sanjit K. Mitra, Oxford University Press
5. Continuous and Discrete-Time Signals and Systems, 2^{nd} Edition, Samir S. Soliman and Mandyam D. Srinath, Prentice-Hall of India.

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CHAPTER 1

SIGNALS